# Book 1 — Lecture 1 — Interlude 1 — Exercise 3

Magnitude Squared and Dot Product

**Prompt.** Show that the magnitude of a vector satisfies

$$\|\mathbf{A}\|^2 = \mathbf{A} \cdot \mathbf{A}.$$

Let

$$\mathbf{A} = (A_1, A_2, \ldots, A_n).$$

**Claim.**

$$\|\mathbf{A}\|^2 = \mathbf{A} \cdot \mathbf{A}.$$

*Proof.* By definition of Euclidean magnitude,

$$\|\mathbf{A}\| = \sqrt{A_1^2 + A_2^2 + \cdots + A_n^2}.$$

Here $\|\mathbf{A}\|$ denotes Euclidean norm[1]. Squaring both sides gives

$$\|\mathbf{A}\|^2 = A_1^2 + A_2^2 + \cdots + A_n^2.$$

By definition of dot product,

$$\mathbf{A} \cdot \mathbf{A} := (A_1, \ldots, A_n) \cdot (A_1, \ldots, A_n) := A_1^2 + A_2^2 + \cdots + A_n^2.$$

The dot product[2] is the standard inner product on $\mathbb{R}^n$[3]. Hence,

$$\|\mathbf{A}\|^2 = \mathbf{A} \cdot \mathbf{A}.$$

□

**Result.** The identity follows directly from component definitions of the Euclidean norm and dot product: both reduce to the same sum of squares.

**Example.** For $\mathbf{A} = (3, -4, 12)$,

$$\mathbf{A} \cdot \mathbf{A} = 3^2 + (-4)^2 + 12^2 = 169, \qquad \|\mathbf{A}\| = 13, \qquad \|\mathbf{A}\|^2 = 169.$$

So the identity holds numerically.

[1] Also called the vector magnitude or length.
[2] For $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$, $\mathbf{u} \cdot \mathbf{v} = \sum_i u_i v_i$.
[3] $\mathbb{R}^n$ is the set of $n$-component real vectors.